View NVMe subsystem controllers

ONTAP 9.13.1 REST API reference

NetApp
February 13, 2024

This PDF was generated from https://docs.netapp.com/us-en/ontap-restapi-9131/ontap/protocols_nvme_subsystem-controllers_endpoint_overview.html on February 13, 2024.
Always check docs.netapp.com for the latest.

Table of Contents

View NVMe subsystem controllers . . . 1
Protocols NVMe subsystem-controllers endpoint overview . . . 1
Retrieve NVMe subsystem controllers . . . 7
Retrieve an NVMe subsystem controller . . . 18

View NVMe subsystem controllers

Protocols NVMe subsystem-controllers endpoint overview

Overview

Non-Volatile Memory Express (NVMe) subsystem controllers represent dynamic connections between hosts and a storage solution.

The NVMe subsystem controllers REST API provides information about connected hosts.

Examples

Retrieving the NVMe subsystem controllers for the entire system

```
# The API:
GET /api/protocols/nvme/subsystem-controllers

# The call:
curl -X GET 'https://<mgmt-ip>/api/protocols/nvme/subsystem-controllers'
-H 'Accept: application/hal+json'

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "f0f5b928-2593-11e9-94c4-00a0989a1c8e",
      "name": "symmcon_fcnvme_vserver_0",
      "_links": {
        "self": {
          "href": "/api/svm/svms/f0f5b928-2593-11e9-94c4-00a0989a1c8e"
        }
      }
    },
    "subsystem": {
      "uuid": "14875240-2594-11e9-abde-00a098984313",
      "name": "symmcon_symmcon_fcnvme_vserver_0_subsystem_0",
      "_links": {
        "self": {
          "href": "/api/protocols/nvme/subsystems/14875240-2594-11e9-abde-
00a098984313"
        }
      }
    },
    "id": "0040h",
```

```
    "_links": {
      "self": {
        "href": "/api/protocols/nvme/subsystem-controllers/14875240-2594-
11e9-abde-00a098984313/0040h"
      }
    }
  },
  {
    "svm": {
      "uuid": "f0f5b928-2593-11e9-94c4-00a0989a1c8e",
      "name": "symmcon_fcnvme_vserver_0",
      "_links": {
        "self": {
          "href": "/api/svm/svms/f0f5b928-2593-11e9-94c4-00a0989a1c8e"
        }
      }
    },
    "subsystem": {
      "uuid": "14875240-2594-11e9-abde-00a098984313",
      "name": "symmcon_symmcon_fcnvme_vserver_0_subsystem_0",
      "_links": {
        "self": {
          "href": "/api/protocols/nvme/subsystems/14875240-2594-11e9-abde-
00a098984313"
        }
      }
    },
    "id": "0041h",
    "_links": {
      "self": {
        "href": "/api/protocols/nvme/subsystem-controllers/14875240-2594-
11e9-abde-00a098984313/0041h"
      }
    }
  },
  {
    "svm": {
      "uuid": "f0f5b928-2593-11e9-94c4-00a0989a1c8e",
      "name": "symmcon_fcnvme_vserver_0",
      "_links": {
        "self": {
          "href": "/api/svm/svms/f0f5b928-2593-11e9-94c4-00a0989a1c8e"
        }
      }
    },
    "subsystem": {
```

```
      "uuid": "1489d0d5-2594-11e9-94c4-00a0989a1c8e",
      "name": "symmcon_symmcon_fcnvme_vserver_0_subsystem_1",
      "_links": {
        "self": {
          "href": "/api/protocols/nvme/subsystems/1489d0d5-2594-11e9-94c4-
00a0989a1c8e"
        }
      }
    },
    "id": "0040h",
    "_links": {
      "self": {
        "href": "/api/protocols/nvme/subsystem-controllers/1489d0d5-2594-
11e9-94c4-00a0989a1c8e/0040h"
      }
    }
  },
  {
    "svm": {
      "uuid": "f0f5b928-2593-11e9-94c4-00a0989a1c8e",
      "name": "symmcon_fcnvme_vserver_0",
      "_links": {
        "self": {
          "href": "/api/svm/svms/f0f5b928-2593-11e9-94c4-00a0989a1c8e"
        }
      }
    },
    "subsystem": {
      "uuid": "1489d0d5-2594-11e9-94c4-00a0989a1c8e",
      "name": "symmcon_symmcon_fcnvme_vserver_0_subsystem_1",
      "_links": {
        "self": {
          "href": "/api/protocols/nvme/subsystems/1489d0d5-2594-11e9-94c4-
00a0989a1c8e"
        }
      }
    },
    "id": "0041h",
    "_links": {
      "self": {
        "href": "/api/protocols/nvme/subsystem-controllers/1489d0d5-2594-
11e9-94c4-00a0989a1c8e/0041h"
      }
    }
  }
],
```

```
"num_records": 4,
"_links": {
  "self": {
    "href": "/api/protocols/nvme/subsystem-controllers"
  }
}
}
```

Retrieving the NVMe subsystem controllers for a specific subsystem

```
# The API:
GET /api/protocols/nvme/subsystem-controllers

# The call:
curl -X GET 'https://<mgmt-ip>/api/protocols/nvme/subsystem-
controllers?subsystem.uuid=14875240-2594-11e9-abde-00a098984313' -H
'Accept: application/hal+json'

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "f0f5b928-2593-11e9-94c4-00a0989a1c8e",
      "name": "symmcon_fcnvme_vserver_0",
      "_links": {
        "self": {
          "href": "/api/svm/svms/f0f5b928-2593-11e9-94c4-00a0989a1c8e"
        }
      }
    },
    "subsystem": {
      "uuid": "14875240-2594-11e9-abde-00a098984313",
      "name": "symmcon_symmcon_fcnvme_vserver_0_subsystem_0",
      "_links": {
        "self": {
          "href": "/api/protocols/nvme/subsystems/14875240-2594-11e9-abde-
00a098984313"
        }
      }
    },
    "id": "0040h",
    "_links": {
      "self": {
```

```
        "href": "/api/protocols/nvme/subsystem-controllers/14875240-2594-
11e9-abde-00a098984313/0040h"
      }
    }
  },
  {
    "svm": {
      "uuid": "f0f5b928-2593-11e9-94c4-00a0989a1c8e",
      "name": "symmcon_fcnvme_vserver_0",
      "_links": {
        "self": {
          "href": "/api/svm/svms/f0f5b928-2593-11e9-94c4-00a0989a1c8e"
        }
      }
    },
    "subsystem": {
      "uuid": "14875240-2594-11e9-abde-00a098984313",
      "name": "symmcon_symmcon_fcnvme_vserver_0_subsystem_0",
      "_links": {
        "self": {
          "href": "/api/protocols/nvme/subsystems/14875240-2594-11e9-abde-
00a098984313"
        }
      }
    },
    "id": "0041h",
    "_links": {
      "self": {
        "href": "/api/protocols/nvme/subsystem-controllers/14875240-2594-
11e9-abde-00a098984313/0041h"
      }
    }
  }
],
"num_records": 2,
"_links": {
  "self": {
    "href": "/api/protocols/nvme/subsystem-controllers/14875240-2594-11e9-
abde-00a098984313"
  }
}
}
```

Retrieving a specific NVMe subsystem controller

```
# The API:
GET /api/protocols/nvme/subsystem-controllers/{subsystem.uuid}/{id}

# The call:
curl -X GET 'https://<mgmt-ip>/api/protocols/nvme/subsystem-
controllers/14875240-2594-11e9-abde-00a098984313/0040h' -H 'Accept:
application/hal+json'

# The response:
{
"svm": {
  "uuid": "f0f5b928-2593-11e9-94c4-00a0989a1c8e",
  "name": "symmcon_fcnvme_vserver_0",
  "_links": {
    "self": {
      "href": "/api/svm/svms/f0f5b928-2593-11e9-94c4-00a0989a1c8e"
    }
  }
},
"subsystem": {
  "uuid": "14875240-2594-11e9-abde-00a098984313",
  "name": "symmcon_symmcon_fcnvme_vserver_0_subsystem_0",
  "_links": {
    "self": {
      "href": "/api/protocols/nvme/subsystems/14875240-2594-11e9-abde-
00a098984313"
    }
  }
},
"id": "0040h",
"interface": {
  "name": "symmcon_lif_fcnvme_symmcon_fcnvme_vserver_0_3a_0",
  "uuid": "fa1c5941-2593-11e9-94c4-00a0989a1c8e",
  "transport_address": "nn-0x200400a0989a1c8d:pn-0x200500a0989a1c8d",
  "_links": {
    "self": {
      "href": "/api/protocols/nvme/interfaces/fa1c5941-2593-11e9-94c4-
00a0989a1c8e"
    }
  }
},
"node": {
  "name": "ssan-8040-94a",
  "uuid": "ebf66f05-2590-11e9-abde-00a098984313",
```

```
    "_links": {
      "self": {
        "href": "/api/cluster/nodes/ebf66f05-2590-11e9-abde-00a098984313"
      }
    }
  },
  "host": {
    "transport_address": "nn-0x20000090fae00806:pn-0x10000090fae00806",
    "nqn": "nqn.2014-08.org.nvmexpress:uuid:c2846cb1-89d2-4020-a3b0-
71ce907b4eef",
    "id": "b8546ca6097349e5b1558dc154fc073b"
  },
  "io_queue": {
    "count": 4,
    "depth": [
      32,
      32,
      32,
      32
    ]
  },
  "admin_queue": {
    "depth": 32
  },
  "dh_hmac_chap": {
    "mode": "none"
  },
  "_links": {
    "self": {
      "href": "/api/protocols/nvme/subsystem-controllers/14875240-2594-11e9-
abde-00a098984313/0040h"
    }
  }
}
```


Retrieve NVMe subsystem controllers

GET `/protocols/nvme/subsystem-controllers`

Introduced In: 9.6

Retrieves NVMe subsystem controllers.

Related ONTAP commands

- `vserver nvme subsystem controller show`

Learn more

- DOC /protocols/nvme/subsystem-controllers

Parameters

Name	Type	In	Required	Description
node.uuid	string	query	False	Filter by node.uuid
node.name	string	query	False	Filter by node.name
io_queue.count	integer	query	False	Filter by io_queue.count
io_queue.depth	integer	query	False	Filter by io_queue.depth
interface.uuid	string	query	False	Filter by interface.uuid
interface.name	string	query	False	Filter by interface.name
interface.transport_address	string	query	False	Filter by interface.transport_address
svm.uuid	string	query	False	Filter by svm.uuid
svm.name	string	query	False	Filter by svm.name
id	string	query	False	Filter by id
host.transport_address	string	query	False	Filter by host.transport_address
host.nqn	string	query	False	Filter by host.nqn • maxLength: 223 • minLength: 1
host.id	string	query	False	Filter by host.id
subsystem.uuid	string	query	False	Filter by subsystem.uuid

Name	Type	In	Required	Description
subsystem.name	string	query	False	Filter by subsystem.name • maxLength: 96 • minLength: 1
admin_queue.depth	integer	query	False	Filter by admin_queue.depth
dh_hmac_chap.mode	string	query	False	Filter by dh_hmac_chap.mode • Introduced in: 9.12
dh_hmac_chap.group_size	string	query	False	Filter by dh_hmac_chap.group_size • Introduced in: 9.12
dh_hmac_chap.hash_function	string	query	False	Filter by dh_hmac_chap.hash_function • Introduced in: 9.12
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Default value: 1 • Max value: 120 • Min value: 0
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
num_records	integer	The number of records in the response.
records	array[nvme_subsystem_controller]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "admin_queue": {
      "depth": 0
    },
    "dh_hmac_chap": {
      "group_size": "none",
      "hash_function": "sha_256",
      "mode": "bidirectional"
    },
    "host": {
      "id": "b8546ca6097349e5b1558dc154fc073b",
      "nqn": "nqn.2014-08.org.nvmexpress:uuid:c2846cb1-89d2-4020-a3b0-
71ce907b4eef",
      "transport_address": "nn-0x20000090fae00806:pn-
0x10000090fae00806"
    },
    "id": "0040h",
    "interface": {
      "name": "lif1",
      "transport_address": "nn-0x200400a0989a1c8d:pn-
0x200500a0989a1c8d",
      "uuid": "fa1c5941-2593-11e9-94c4-00a0989a1c8e"
    },
    "io_queue": {
      "count": 0,
      "depth": {
      }
    },
    "node": {
```

```
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "node1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "subsystem": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "subsystem1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    }
  }
}
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
next	href	
self	href	

_links

Name	Type	Description
self	href	

admin_queue

Name	Type	Description
depth	integer	The depth of the admin queue for the controller.

dh_hmac_chap

A container for properties of the NVMe in-band authentication DH-HMAC-CHAP protocol used by the the host connection to the controller.

Name	Type	Description
group_size	string	The Diffie-Hellman group size used for NVMe in-band authentication.
hash_function	string	The hash function used for NVMe in-band authentication.

Name	Type	Description
mode	string	The NVMe in-band authentication mode used for the host connection. When set to: • none: Neither the host nor controller was authenticated. • unidirectional: The controller authenticated the host. • bidirectional: The controller authenticated the host and the host authenticated the controller.

host

Properties of the connected host.

Name	Type	Description
id	string	The host identifier registered with the controller.
nqn	string	The NVMe qualified name of the host.
transport_address	string	The transport address of the host.

interface

The logical interface through which the host is connected.

Name	Type	Description
name	string	The name of the logical interface.
transport_address	string	The transport address of the logical interface.
uuid	string	The unique identifier of the logical interface.

io_queue

Properties of the I/O queues available to the controller.

Name	Type	Description
count	integer	The number of I/O queues available to the controller.
depth	array[integer]	The depths of the I/O queues.

node

Name	Type	Description
_links	_links	
name	string	
uuid	string	

subsystem

Name	Type	Description
_links	_links	
name	string	The name of the NVMe subsystem.
uuid	string	The unique identifier of the NVMe subsystem.

svm

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

nvme_subsystem_controller

A Non-Volatile Memory Express (NVMe) subsystem controller represents a connection between a host and a storage solution.

An NVMe subsystem controller is identified by the NVMe subsystem UUID and the controller ID.

Name	Type	Description
_links	_links	
admin_queue	admin_queue	

Name	Type	Description
dh_hmac_chap	dh_hmac_chap	A container for properties of the NVMe in-band authentication DH-HMAC-CHAP protocol used by the the host connection to the controller.
host	host	Properties of the connected host.
id	string	The identifier of the subsystem controller. This field consists of 4 zero-filled hexadecimal digits followed by an 'h'.
interface	interface	The logical interface through which the host is connected.
io_queue	io_queue	Properties of the I/O queues available to the controller.
node	node	
subsystem	subsystem	
svm	svm	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve an NVMe subsystem controller

GET `/protocols/nvme/subsystem-controllers/{subsystem.uuid}/{id}`

Introduced In: 9.6

Retrieves an NVMe subsystem controller.

Related ONTAP commands

- `vserver nvme subsystem controller show`

Learn more

- DOC /protocols/nvme/subsystem-controllers

Parameters

Name	Type	In	Required	Description
subsystem.uuid	string	path	True	The unique identifier of the NVMe subsystem.
id	string	path	True	The unique identifier of the NVMe subsystem controller.
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
admin_queue	admin_queue	
dh_hmac_chap	dh_hmac_chap	A container for properties of the NVMe in-band authentication DH-HMAC-CHAP protocol used by the the host connection to the controller.
host	host	Properties of the connected host.

Name	Type	Description
id	string	The identifier of the subsystem controller. This field consists of 4 zero-filled hexadecimal digits followed by an 'h'.
interface	interface	The logical interface through which the host is connected.
io_queue	io_queue	Properties of the I/O queues available to the controller.
node	node	
subsystem	subsystem	
svm	svm	

Example response


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "admin_queue": {
    "depth": 0
  },
  "dh_hmac_chap": {
    "group_size": "none",
    "hash_function": "sha_256",
    "mode": "bidirectional"
  },
  "host": {
    "id": "b8546ca6097349e5b1558dc154fc073b",
    "nqn": "nqn.2014-08.org.nvmexpress:uuid:c2846cb1-89d2-4020-a3b0-
71ce907b4eef",
    "transport_address": "nn-0x20000090fae00806:pn-0x10000090fae00806"
  },
  "id": "0040h",
  "interface": {
    "name": "lif1",
    "transport_address": "nn-0x200400a0989a1c8d:pn-0x200500a0989a1c8d",
    "uuid": "fa1c5941-2593-11e9-94c4-00a0989a1c8e"
  },
  "io_queue": {
    "count": 0,
    "depth": {
    }
  },
  "node": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "node1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "subsystem": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
```

```
      }
    },
    "name": "subsystem1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```


Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
72090001	The supplied subsystem identifier does not exist.

Name	Type	Description
error	error	

Example error


```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

admin_queue

Name	Type	Description
depth	integer	The depth of the admin queue for the controller.

dh_hmac_chap

A container for properties of the NVMe in-band authentication DH-HMAC-CHAP protocol used by the the host connection to the controller.

Name	Type	Description
group_size	string	The Diffie-Hellman group size used for NVMe in-band authentication.
hash_function	string	The hash function used for NVMe in-band authentication.
mode	string	The NVMe in-band authentication mode used for the host connection. When set to: • none: Neither the host nor controller was authenticated. • unidirectional: The controller authenticated the host. • bidirectional: The controller authenticated the host and the host authenticated the controller.

host

Properties of the connected host.

Name	Type	Description
id	string	The host identifier registered with the controller.
nqn	string	The NVMe qualified name of the host.
transport_address	string	The transport address of the host.

interface

The logical interface through which the host is connected.

Name	Type	Description
name	string	The name of the logical interface.
transport_address	string	The transport address of the logical interface.
uuid	string	The unique identifier of the logical interface.

io_queue

Properties of the I/O queues available to the controller.

Name	Type	Description
count	integer	The number of I/O queues available to the controller.
depth	array[integer]	The depths of the I/O queues.

node

Name	Type	Description
_links	_links	
name	string	
uuid	string	

subsystem

Name	Type	Description
_links	_links	

Name	Type	Description
name	string	The name of the NVMe subsystem.
uuid	string	The unique identifier of the NVMe subsystem.

svm

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Copyright information

Copyright © 2024 NetApp, Inc. All Rights Reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means—graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system—without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

LIMITED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (b)(3) of the Rights in Technical Data -Noncommercial Items at DFARS 252.227-7013 (FEB 2014) and FAR 52.227-19 (DEC 2007).

Data contained herein pertains to a commercial product and/or commercial service (as defined in FAR 2.101) and is proprietary to NetApp, Inc. All NetApp technical data and computer software provided under this Agreement is commercial in nature and developed solely at private expense. The U.S. Government has a non-exclusive, non-transferrable, nonsublicensable, worldwide, limited irrevocable license to use the Data only in connection with and in support of the U.S. Government contract under which the Data was delivered. Except as provided herein, the Data may not be used, disclosed, reproduced, modified, performed, or displayed without the prior written approval of NetApp, Inc. United States Government license rights for the Department of Defense are limited to those rights identified in DFARS clause 252.227-7015(b) (FEB 2014).

Trademark information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.